A Version Assist for Digital Scholarly Editions

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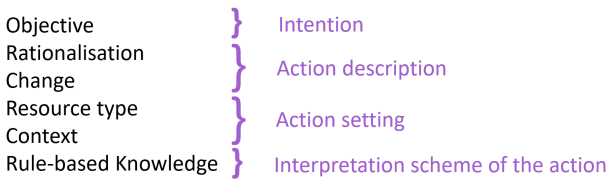
When we talk about dynamic digital resources, we are referring to temporary versions that represent the current state of the resource for a specific period. By documenting these versions and the changes they undergo, we can examine and analyse the version history of the resource. In the context of scholarly research resources, this enables us to draw insights regarding the research perspective and the contextualisation of the research subject.

The necessity for transparency and clarity in change processes and their underlying motives is especially significant in the field of humanities research. This is due to the fact that research subjects can be examined from various perspectives and can never be fully, definitively, or unambiguously described. Unlike other scientific disciplines, the knowledge produced in the humanities tends to be less cumulative. Given these factors, it is essential to develop and work with different versions.

The inclination to create versions is a defining feature not just of humanities research, but especially of digital research. Because data and models are inherently characterized by perspectivity, provisionality, underdetermination, and ambiguity. Within the research community, the ontological definitions of digital objects and texts must be continually redefined. This ongoing process is reflected in the creation of various versions. Additionally, there is a prevailing expectation in the scientific community to publish findings promptly, which aims to enhance the speed of scientific communication and ensure that the research process remains transparent and accessible. This commitment, along with the technical requirement to maintain, adjust, and refresh the ongoing usability of digital research resources—such as digital editions, collections, and archives, as well as data and programming interfaces—increases the need for a systematic approach to versioning (cp. Brown et al. 2014, Pierazzo 2014 pp. 198-201, Broyles 2020).

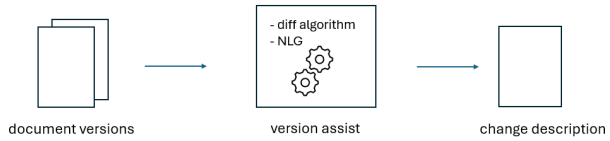
To facilitate the description of versions and the creation of a version history, this article proposes a version assist system for digital editions. The version assist returns automatically generated change descriptions of changed resources that are comprehensible because each change is described as a purposeful, rule-based and contextualised action. Comprehensible descriptions are guaranteed because they are based on the model of Intentional Versioning (Bürgermeister 2025).



Model of Intentional Versioning

The Intentional Versioning description model allows for the clear representation of changes driven by various intentions. Each change is depicted within its specific context of action. Every actual change is articulated as an action at least twice: once as a means to an end and once as the end itself. This means-end relationship becomes clearer when it is linked to a defined future objective. The desired outcome, in turn, serves as the focus of the intention, making it evident.

The ordered nature of an intentional action is reconstructed by the entities Change, Rationalisation and Objective. These actions are situated within their context through the entities Resource type and Context. To execute and articulate an action in a specific scenario, certain knowledge is necessary, which is encapsulated by the entity Rule-based knowledge.



Version assist

The most recent release version and the current version of an XML document serve as the input for the version assist (Fig. 2). The system analyses these document versions by employing a tree-diff algorithm to identify differences and generate a delta. This delta is then translated into intentional actions through a simple Natural Language Generator (NLG). This system operates as a data-to-text solution, producing textual descriptions based on formalized knowledge associated with the edition project. This knowledge encompasses schema files (such as the TEI-schema), which document the meanings of editing and coding rules, along

with other formal definitions outlined in the edition plan. The resulting change descriptions can be modelled as a graph along with other provenance metadata and linked to the current document. Additionally, these automatically generated descriptions can be utilized to create a version history within the user interface of the digital edition. Figure 3 below visualises a possible version history for a resource.

Version history

Resource: Notizbuch (Nov 1967-Mär 1977)

Date: 2024-07-29

Context: page 76, //div/pb[@n=,76‘]/lb[@n=,11‘]

Rationalisation: Editorial intervention, spelling corrected

Change:

```
<add place="above"><unclear reason="faded" cert="low">Ober
</unclear></add><unclear cert="medium">Li</unclear>ppe
```

```
<choice>
  <sic><add place="above"><unclear reason="faded" cert="low">Ober
  </unclear></add><unclear cert="medium">Li</unclear>ppe</sic>
  <corr><add place="above"><unclear reason="faded" cert="low">Ober-
  </unclear></add><unclear cert="medium">Li</unclear>ppe</corr>
</choice>
```

Date: 2024-03-02

Context: page 99, //div/pb[@n=,99‘]/lb[@n=,25‘]

Rationalisation: Transcription corrected, editorial intervention commented as uncertain

Change:

```
<seg type="note">Ein Holler<hi rend="underlined">baum</hi>
  <pc type="single">(</pc>an dem wieder andre Rauten sind
  <supplied>)</supplied></seg>

<seg type="note">Ein Holler<hi rend="underlined">baum</hi>
  <pc type="single">(</pc>an dem wieder andre Ranken sind
  <supplied cert="low">)</supplied></seg>
```

Visualisation of a version history (1).

The version history in Figure 3 shows two entries. They represent versions of a particular resource. Each entry has a date and information on the context of the change, the description of the purpose of the change and the means used.

All description elements should be clear and comprehensible at the presentation level of the digital edition, allowing users to quickly evaluate the version and determine the relevance of the changes to them. As illustrated in the example entries (Fig. 3), the page number serves as a helpful reference point in digital texts.

Figure 4 demonstrates another example entry in a version history that describes a change that affects neither the content of the resource nor the meaning level of the modelling. However, the intention of these changes can be reconstructed thanks to the model-components Change, Rationalisation and Objective.

Objective: Automated data maintainance, data consistency

Context: Commentary page 4, //div/pb[@n=,4‘]

Rationalisation: Hash values replaces

Change:

```
<ref xml:id="1ae5c0e84c30df4bc7405d8fcba9f2cc9354aecb"
  corresp="#db825a11803dbba905743281edcdc8cd40a0643e">
  <date when="1927-07-30">30 Juli</date>
</ref>
<note xml:id="db825a11803dbba905743281edcdc8cd40a0643e"
  corresp="#1ae5c0e84c30df4bc7405d8fcba9f2cc9354aecb"
  >Das war der Tag der Eröffnung der Salzburger Festspiele, wo [...]</note>

<ref xml:id="r516b416b9f8e8d388cf38ef715a77e6c"
  corresp="#k18b612b43d5996fd7a34454ba3604a04">
  <date when="1927-07-30">30 Juli</date>
</ref>
<note xml:id="k18b612b43d5996fd7a34454ba3604a04"
  corresp="#r516b416b9f8e8d388cf38ef715a77e6c"
  >Das war der Tag der Eröffnung der Salzburger Festspiele, wo [...]</note>
```

Visualisation of a version history (2).

The Intentional Versioning Model allows for the tracking of intricate research processes while also enabling the selection of important versions. This approach fosters trust, as the actions leading to changes can be validated and justified. In this context, research in the humanities can thrive, allowing for the continuous revision and reinterpretation of its subjects.

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